

LAB 08

Class Task 1:

```
classtask1.cpp
#include<iostream>
using namespace std;
class Student{
    int mathMarks;
    int engMarks;
    int CompMarks;
public:
    void setMarks(int a,int b,int c){ if((a<100 && a>0) && (b<100 && b>0) && (c<100 && c>0)){
        mathMarks=a;engMarks=b;CompMarks=c;
    }}
    friend float calcAvg(Student obj);
};
float calcAvg(Student obj){
    float avg=(obj.mathMarks+obj.engMarks+obj.CompMarks)/3.0;
    return avg;
}
int main(){
    Student s;int a;int b;int c;
    cout<<"Enter student marks: ";
    cin>>a;cin>>b;cin>>c;
    s.setMarks(a,b,c);
    cout<<"Average: "<<calcAvg(s)<<endl;
}
```

Output:

```
pp -o classtask1 && "/home/rumaisa-
Enter student marks: 45
43
46
Average: 44.6667
rumaisa-abbasi@abduallah-muaz-ThinkF
```

ClassTask 2:

```
#include<iostream>
using namespace std;
class Library{
    bool availibility;
    string title;
public:
    Library(){ availibility=1;
    }
    friend class Librarian;
};
class Librarian{
public:
    void issueBook(Library &obj){
        if(obj.availibility){
            cout<<"Book Issued.."<<endl; obj.availibility=0;
        }
    }
    void returnBook(Library &obj){
        if(obj.availibility==0){
            cout<<"Book Returned.."<<endl; obj.availibility=1;
        }
    }
};
int main(){
    Library book;
    Librarian l;
    l.issueBook(book);
    l.returnBook(book);
}
```

Output:

```
● rumaisa-abbasi@abduallah-muaz-ThinkPa
pp -o classtask2 && "/home/rumaisa-a
Book Issued..
Book Returned..
○ rumaisa-abbasi@abduallah-muaz-ThinkPa
```

Class Task 3:

```
#include<iostream>
using namespace std;
class Person{
protected:
    string name;
public:
    Person(){ }
    Person(string n):name(n){}
};
class Employee: virtual public Person{
protected:
    int ID;
public:
    Employee(int id):ID(id),Person(){}
};
class Student: virtual public Person{
protected:
    float GPA;
public:
    Student(float gpa):GPA(gpa),Person(){}
};
class TeachingAssistant:public Employee,public Student{
public:
    TeachingAssistant(string n,int id,float gpa):Person(n),Employee(id),Student(gpa){}
    void display(){
        cout<<"Name: "<<name<<"\nID: "<<ID<<"\nGPA: "<<GPA<<endl;
    }
};
int main(){
    TeachingAssistant ta("Ali",1001,3.7);
    ta.display();
}
```

Output:

```
pp -o classtask3 && "/home/rumaisa-abbasi/De
Name: Ali
ID: 1001
GPA: 3.7
o rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~
```

Lab Task 1:

```
ltask1.cpp
1  #include<iostream>
2  using namespace std;
3  class Rectangle{
4      int length;
5      int width;
6  public:
7      Rectangle(int l,int w){
8          if(l>0 && w>0){length=l;width=w;}
9          else{length=0;width=0;}}
10     friend void calc(Rectangle obj);
11 };
12 void calc(Rectangle obj){
13     cout<<"Area: "<<obj.length*obj.width<<"\nPerimeter: "<<(2*obj.length)+(2*obj.width)<<endl;
14 }
15 int main(){
16     int l,int p;
17     cout<<"Enter length and perimeter: ";
18     cin>>l;cin>>p;
19     Rectangle r(l,p);
20     calc(r);
21 }
```

Output:

```
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab
/" && g++ ltask1.cpp -o ltask1 && "/home/rumaisa-abbasi/
Enter length and width: 4
2
Area: 8
Perimeter: 12
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop/Lab
```

Lab Task 2:

```
#/Desktop/Lab 08/ltask2.cpp
using namespace std;
class Engine{
    int horsepower;
public:
    Engine(int h):horsepower(h){}
    friend class Mechanic;
};
class Mechanic{
public:
    void showHP(Engine e){ cout<<"Horse Power: "<<e.horsepower<<endl;}
    void upgradeHP(Engine& e,int a){ if( e.horsepower+a>=100 && e.horsepower+a<=1000)
        {e.horsepower+=a;}}
};
int main(){
    Engine e(520);
    Mechanic m;
    m.showHP(e);
    m.upgradeHP(e,40);
    m.showHP(e);
}
```

Output:

```
op/Lab 08/"ltask2
● rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop
/" && g++ ltask2.cpp -o ltask2 && "/home/rumaisa-a
Horse Power: 520
Horse Power: 560
○ rumaisa-abbasi@abdullah-muaz-ThinkPad-L390:~/Desktop
```

Lab Task 3:

```
#include <iostream>
using namespace std;
class Person {
private:
    int ID;
    string name;
public:
    Person(int i, string n) : ID(i), name(n) {}
    void displayPerson() {
        cout << "ID: " << ID << ", Name: " << name << endl; }
};

class Student : public Person {
protected:
    float GPA;
public:
    Student(int i, string n, float g) : Person(i,n), GPA(g) {}
    void displayStudent() {
        displayPerson();
        cout << "GPA: " << GPA << endl; }
};

class Employee : public Person {
protected:
    float Salary;
public:
    Employee(int i, string n, float s) : Person(i,n), Salary(s) {}
    void displayEmployee() {
        displayPerson();
        cout << "Salary: " << Salary << endl; }
};

class TeachingAssistant : public Student, public Employee {
public:
    TeachingAssistant(int i1, string n1, float g, int i2, string n2, float s)
        : Student(i1,n1,g), Employee(i2,n2,s) {}
    void displayTA() {
        Student::displayStudent();
        Employee::displayEmployee(); }
};

int main() {
    TeachingAssistant ta(1,"Ali",3.8, 2,"Ali", 5000);
    ta.displayTA();
    return 0;
}
```

Output:

```
rumaisa-abbasi@abdulla:~/C++$ g++ ltask3.cpp -o ltask3
./ltask3
ID: 1, Name: Ali
GPA: 3.8
ID: 2, Name: Ali
Salary: 5000
rumaisa-abbasi@abdulla:~/C++$
```

